Modeling Neutron Stars
Lecture-2

Chiranjib Mondal
LPC, Université Caen Normandie
https://sites.google.com/view/nucleartheory/home

UNIVERSITÉ CAEN NORMANDIE

Nuclear Theory Group
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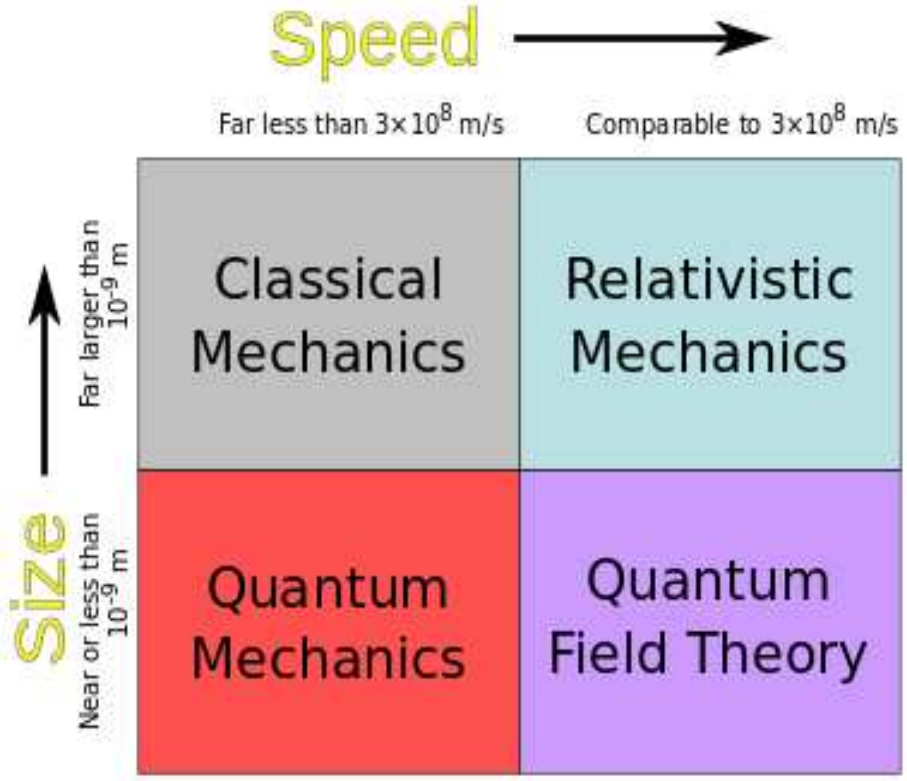
Contents:

Elementary Field Theory

- Why field theory?
- Recapitulation of Special Theory of Relativity
- Maxwell's equations
- Quick review on Particle Mechanics

Different types of dynamics in nature



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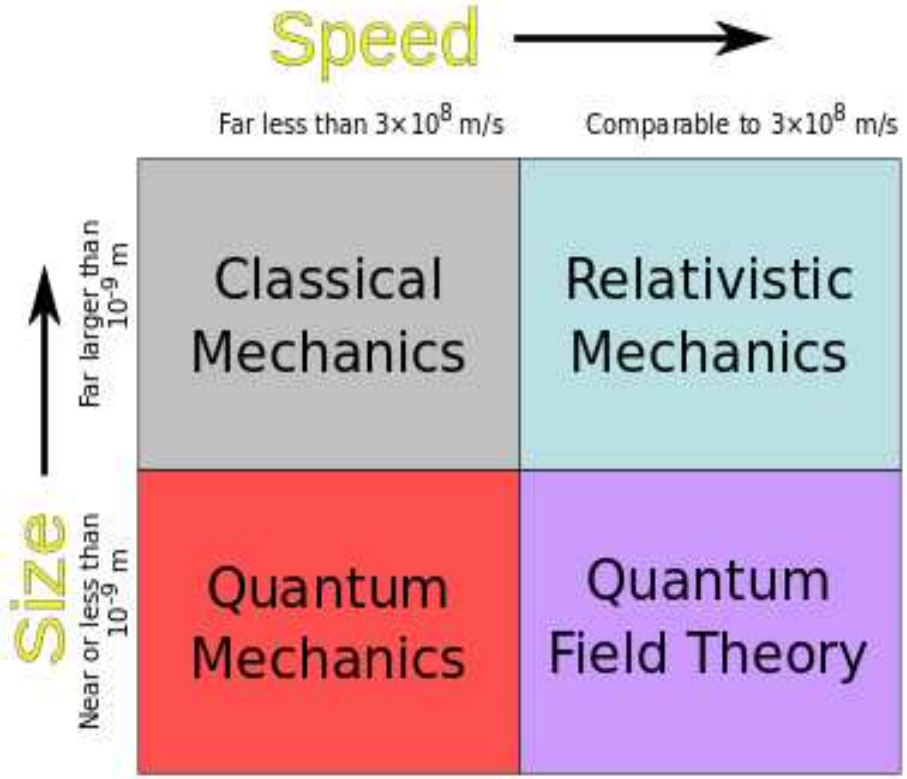

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- Interaction not instantaneous (no potential)
- Naturally incorporates STR
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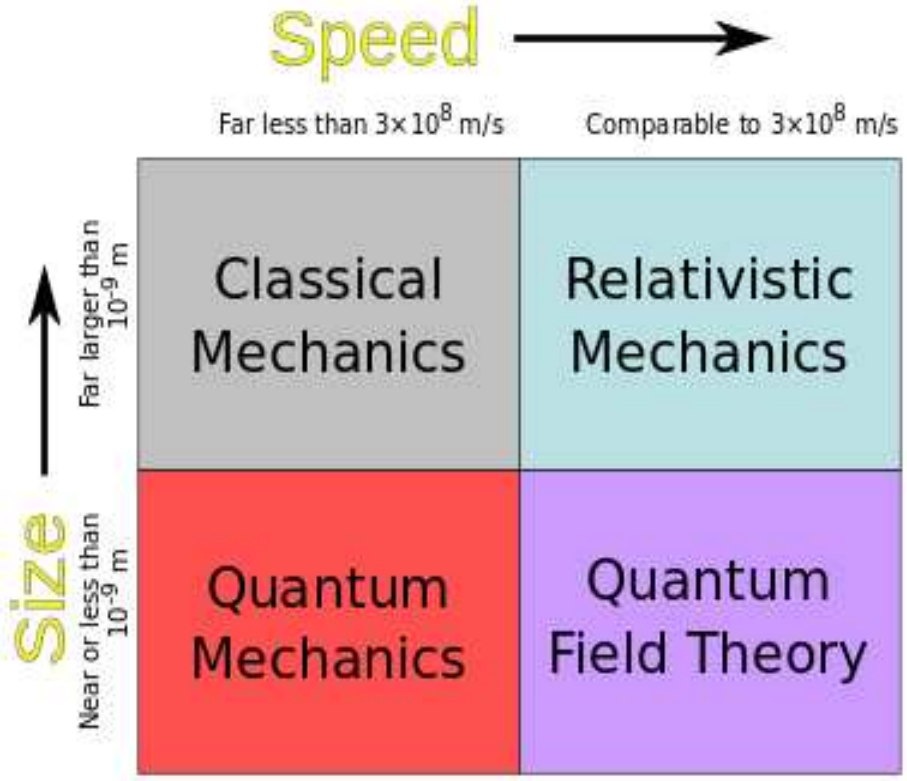

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Goal to learn:

- Transformations and Invariances
- Euler-Lagrange Equations
- Energy-Momentum Tensor

Motivation to do field theory

- Motion of a particle under a potential is ad-hoc.

$$-\frac{GM}{r} \qquad \text{or} \qquad \frac{1}{4\pi\varepsilon_0}\frac{Q}{r}$$

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Goal: To build EoS in in a relativistic model

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Space-time notations

- Greek indices $\mu, \nu, \lambda, \cdots \in 1, 2, 3, 4 \rightarrow$ components of 4-vector.
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- The *"metric tensor"* $g^{\mu\nu}$ is given by

$$g^{\mu\nu} = \mathrm{diag}\{1, -1, -1, -1\}$$

Properties of *metric tensor*

Caution: Repeated indices are summed

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- The Kronecker delta follows:

$$\delta^\mu_\lambda = \begin{cases} 1 \text{ if } \mu = \lambda \\ 0 \text{ if } \mu \neq \lambda \end{cases}$$

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- **Problem 1: Show that the Lorentz transformation rule for x_μ is given by: $x_\mu \to x'_\mu = \Lambda_\mu{}^\nu x_\nu$.**
- **Problem 2: If the Lorentz transformation equations are given by $t' = \gamma(t - vx/c^2)$, $x' = \gamma(x - vt)$, $y' = y$ and $z' = z$, where, $\gamma = (1 - v^2/c^2)^{-1/2}$, show that $x'^\mu x'_\mu$ remains invariant under this transformation. Show that these transformations can be written in the form $x^\mu \to x'^\mu = \Lambda^\mu{}_\nu x^\nu$ with $\Lambda_{\mu\nu} = g_{\mu\nu} + \omega_{\mu\nu}$ for infinitesimal values of v, where, $\omega_{\mu\nu} = -\omega_{\nu\mu}$.**

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\begin{pmatrix} ct' \\ x' \\ y' \\ z' \end{pmatrix} = \begin{pmatrix} \gamma & -\gamma\frac{v}{c} & 0 & 0 \\ -\gamma\frac{v}{c} & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} ct \\ x \\ y \\ z \end{pmatrix}
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Some other physical properties

- Scalar product two vectors A^μ and B^μ is written as:

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- **Problem 3: Convince yourself that at non-relativistic limit $\left(\frac{v}{c} \to 0\right)$ this leads to $E = mc^2$.**
- **Problem 4: Write the Maxwell's equations in 4-vector notation.**

Maxwell's equations

4-vector form!

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$$

Natural Units

$\hbar = c = 1$

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Quantity	Dimension	Conversion factor
Mass	$[\mathrm{M}]$	$1/c^2$
Length	$[\mathrm{M}]^{-1}$	$\hbar c$
Time	$[\mathrm{M}]^{-1}$	$\hbar$
Energy	$[\mathrm{M}]$	1
Momentum	$[\mathrm{M}]$	$1/c$
Force	$[\mathrm{M}]^2$	$1/\hbar c$
Action	$[\mathrm{M}]^0$	$\hbar$
Electric Charge	$[\mathrm{M}]^0$	$\sqrt{\hbar c}$

Action Principle

Particle mechanics

- Theories of classical physics can be described by starting from the principle of least action. A set of coordinates $q_r(t)$ and their time derivatives $\dot{q}_r(t)$ can describe the configuration and the evolution of the system.

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- **Principle of least action:** Among all trajectories that join $q_r(t_1)$ and $q_r(t_2)$, the system follows the one for which $\mathscr{A}$ is stationary.

Action Principle

Euler-Lagrange Equations

Action Principle

Euler-Lagrange Equations

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Legendre Transformation

- For two functions $\mathrm{F}(\mathbf{u}, \mathbf{w})$ and $\mathrm{G}(\mathbf{v}, \mathbf{w})$ where $\mathbf{v} = \boldsymbol{\nabla}_{\mathrm{u}}\mathrm{F}(\mathbf{u}, \mathbf{w})$,

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- **Problem 7: Write the Lagrangian for a free falling particle under gravity. Show that Euler-Lagrangian equation leads to Newtonian equation of motion.**

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- **Problem 8: Derive the Hamilton's equation of motion starting from the above relation. Show that $[\boldsymbol{L}_i, \boldsymbol{L}_j]_{\mathrm{P}} = \varepsilon_{ijk}\boldsymbol{L}_k$, where, $\boldsymbol{L} = \boldsymbol{r} \times \boldsymbol{p}$ is the angular momentum.**